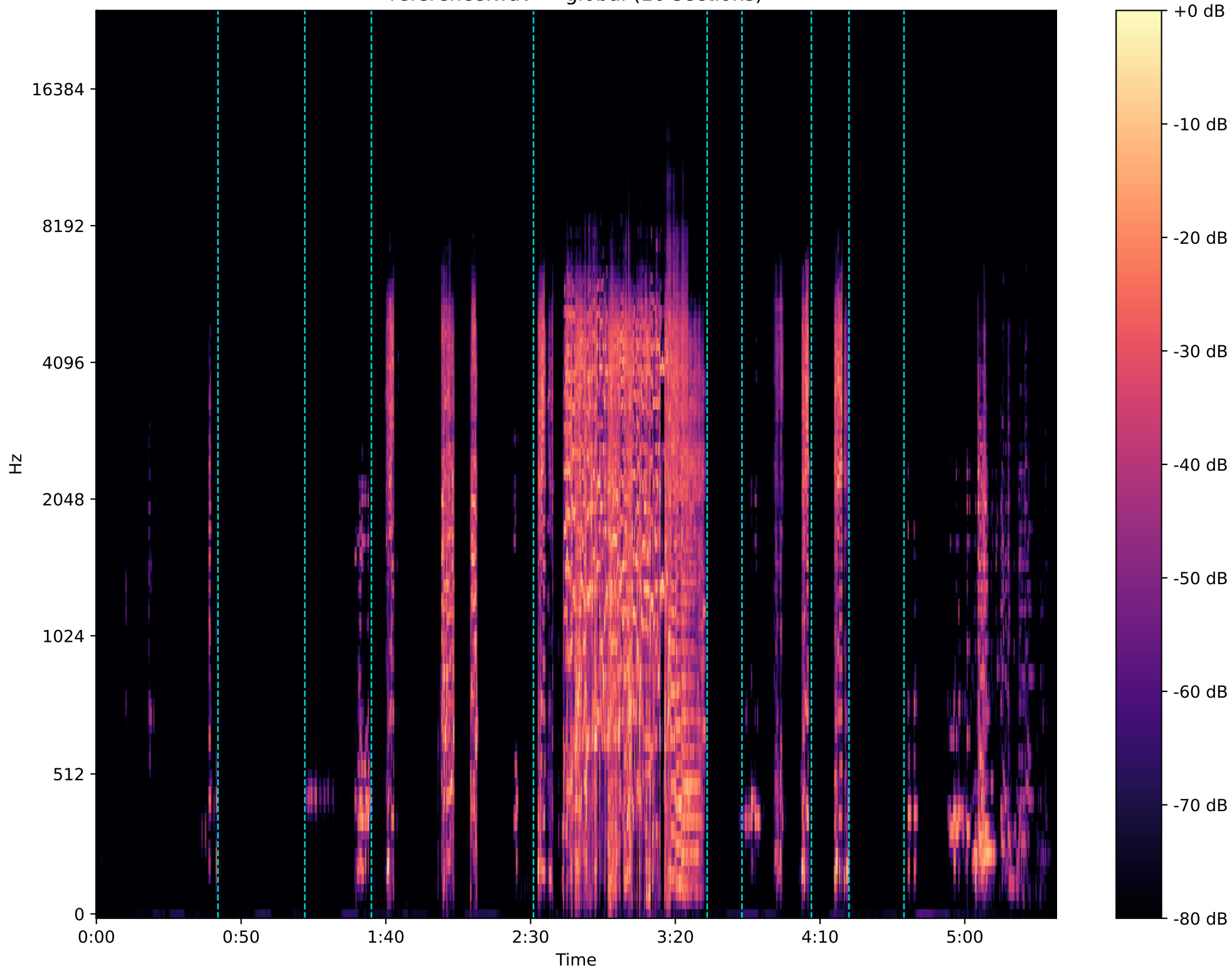


Tone Analysis — reference.wav

Source	/Users/joao.faria/Projetos/github.com/jpfaria/tone-analyzer/tones/metallica-e
SHA-256	7bed48d4f770615597357b4aab565ac5fc8d9832a464fe8e4e0d9b95ae29c4a5
Sample rate	48000 Hz
Channels	2
Duration	331.56 s
Schema	v4
LUFS (int.)	-22.1 LUFS
Peak	-8.3 dB
Stereo width	—
Sections	10 (see following pages)

reference.wav — global (10 sections)



section_0 · 0.00 s - 42.00 s · clean

Labels tone_profile = clean dynamics_profile = sparse presence = background

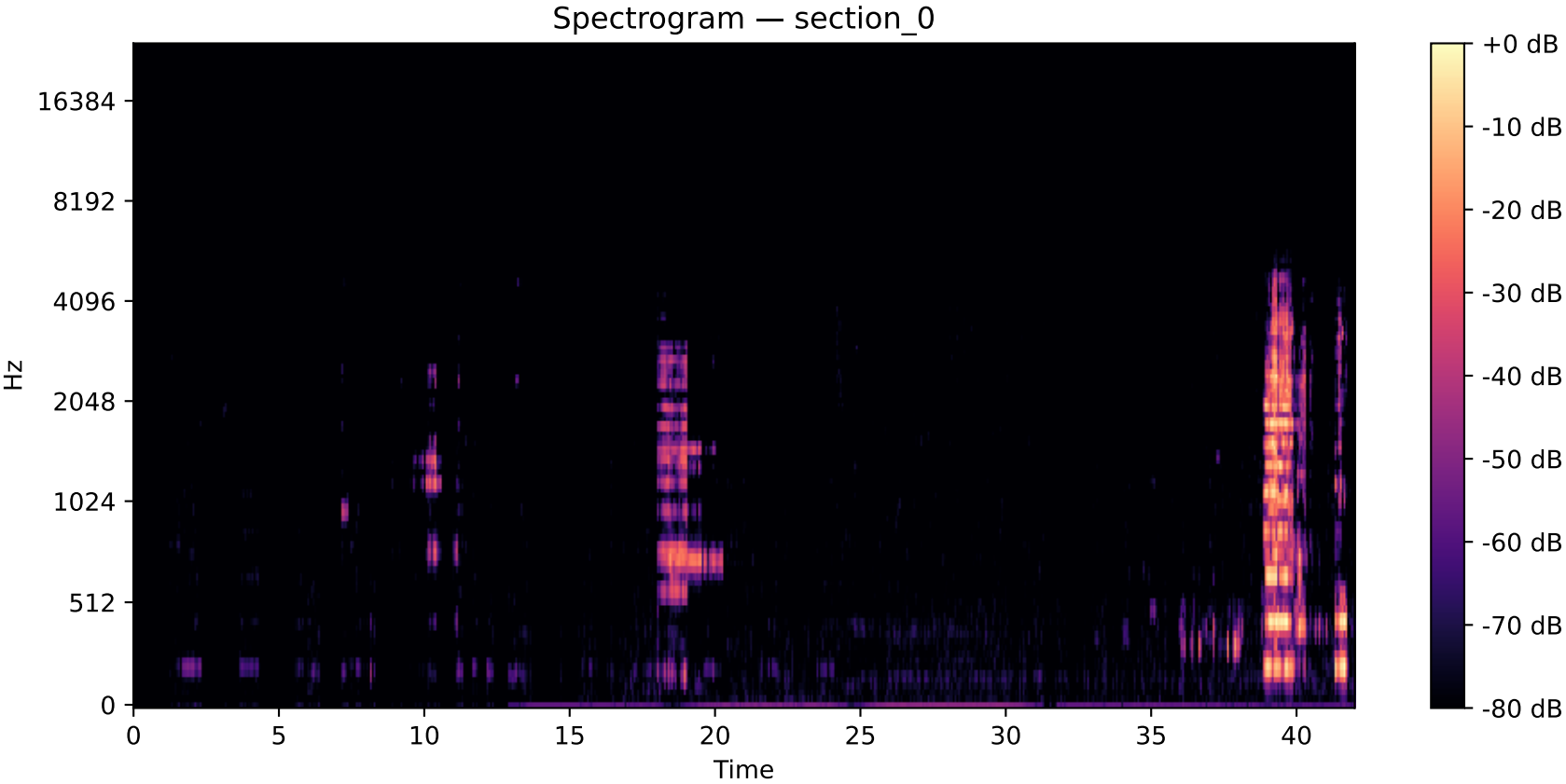
Loudness RMS = -52.2 dB Peak = -23.5 dB Crest = 28.7 dB

Spectrum Centroid = 1642 Hz Rolloff (85%) = 1992 Hz Flatness = 0.001

Distortion THD $\approx$ 0.0 % odd/even = +0.0 dB gain_character = clean (conf 1.00)

Time FX RT60 = 3.15 s delay = off (–, fb –) mod = off (–, depth –)

Band energy (dB) 80 Hz: -100.6 160 Hz: -61.8 320 Hz: -58.8 640 Hz: -60.6
1280 Hz: -58.2 2560 Hz: -67.8 5120 Hz: -113.1 10240 Hz: -113.0



section_1 · 42.00 s - 72.00 s · clean

Labels tone_profile = clean dynamics_profile = rhythmic presence = background

Loudness RMS = -88.3 dB Peak = -68.0 dB Crest = 20.2 dB

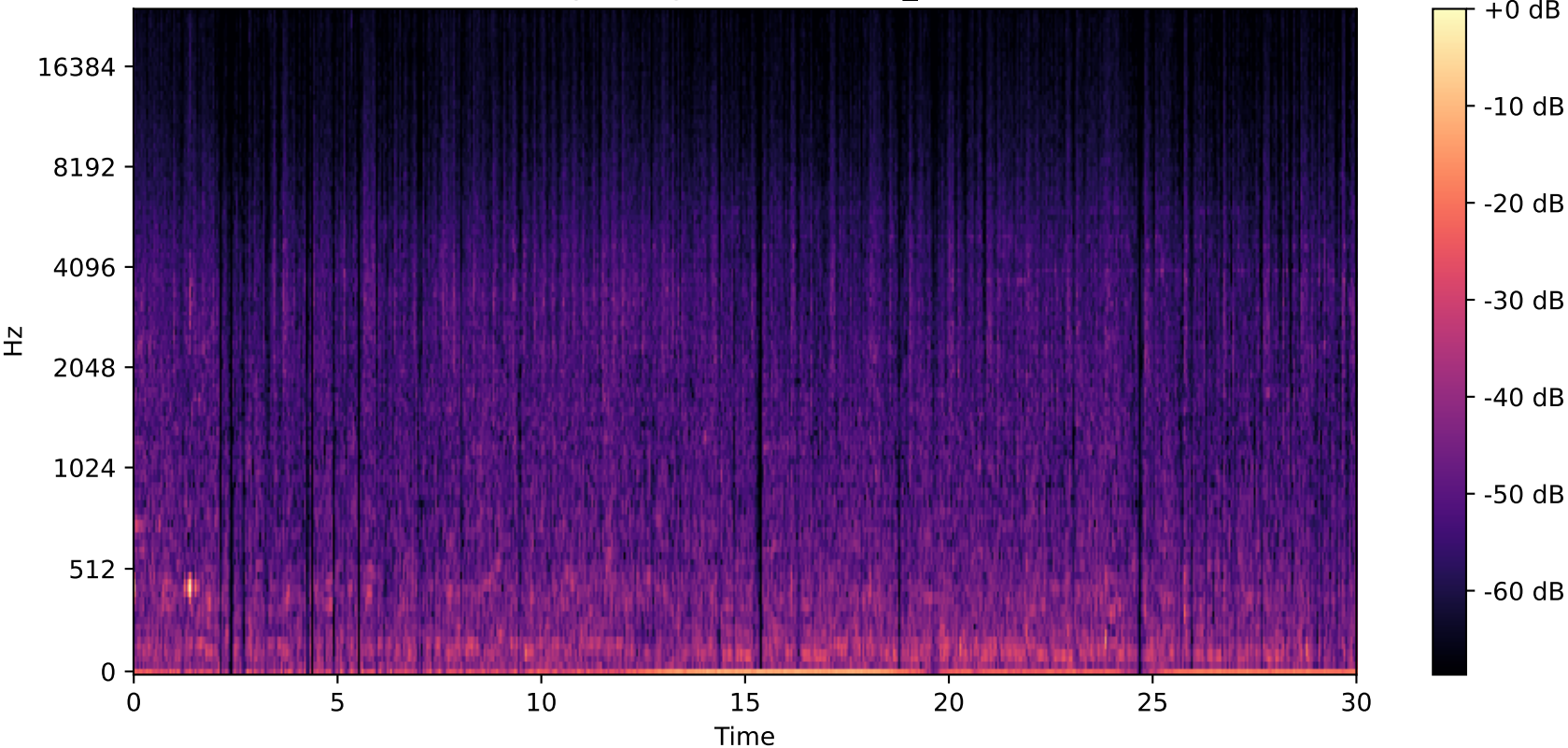
Spectrum Centroid = 5597 Hz Rolloff (85%) = 23 Hz Flatness = 0.389

Distortion THD $\approx$ 31.5 % odd/even = -0.4 dB gain_character = clean (conf 0.20)

Time FX RT60 = - s delay = off (-, fb -) mod = off (-, depth -)

Band energy (dB) 80 Hz: -107.1 160 Hz: -108.3 320 Hz: -103.2 640 Hz: -112.0
1280 Hz: -111.3 2560 Hz: -110.3 5120 Hz: -113.1 10240 Hz: -114.6

Spectrogram — section_1



section_2 · 72.00 s - 95.00 s · crunch

Labels tone_profile = crunch dynamics_profile = sparse presence = background

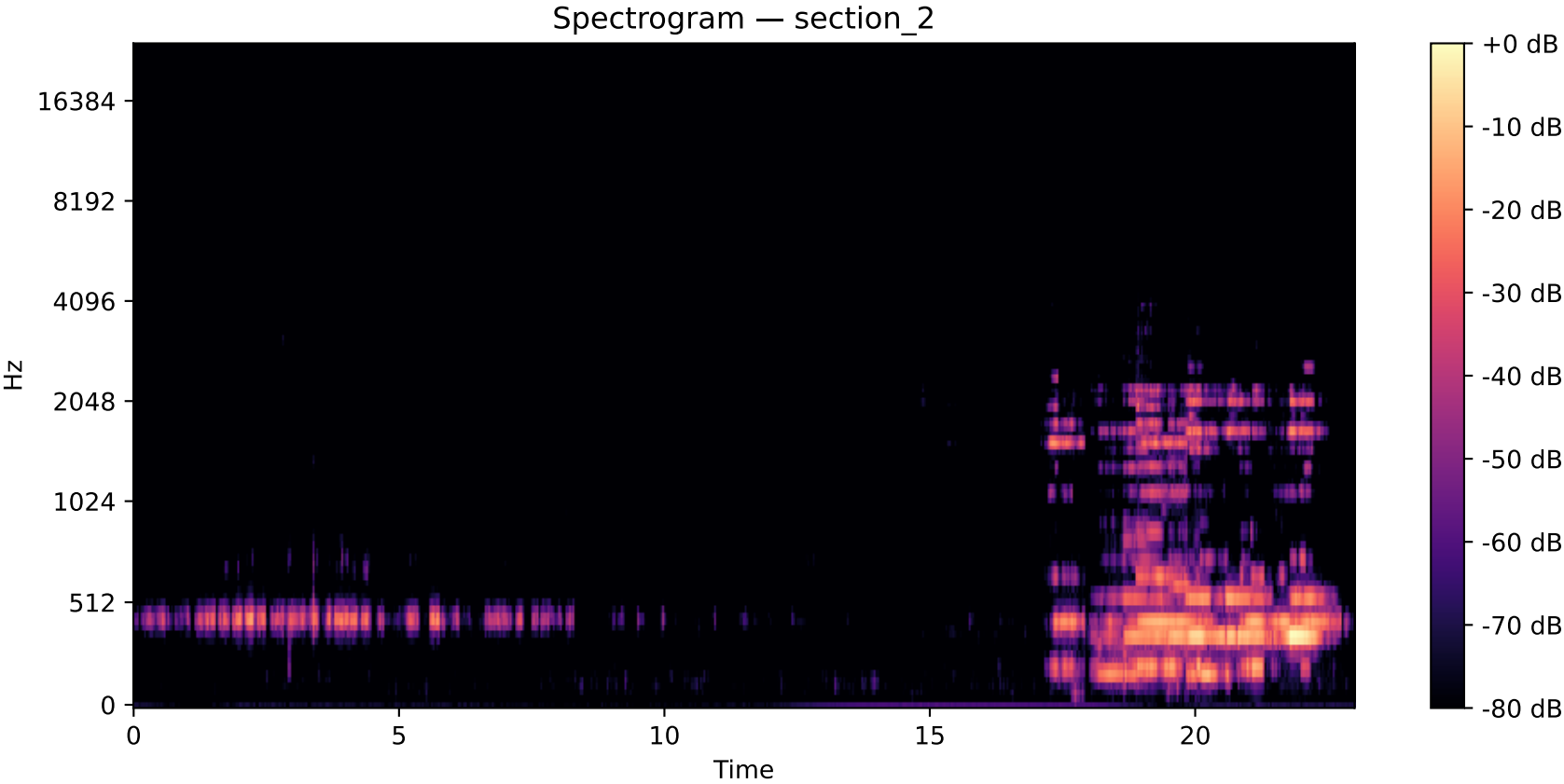
Loudness RMS = -38.4 dB Peak = -17.1 dB Crest = 21.3 dB

Spectrum Centroid = 742 Hz Rolloff (85%) = 445 Hz Flatness = 0.000

Distortion THD $\approx$ 6.8 % odd/even = -1.9 dB gain_character = crunch (conf 0.65)

Time FX RT60 = 1.43 s delay = off (–, fb –) mod = off (–, depth –)

Band energy (dB) 80 Hz: -80.1 160 Hz: -50.6 320 Hz: -40.2 640 Hz: -62.0
1280 Hz: -54.5 2560 Hz: -82.5 5120 Hz: -109.7 10240 Hz: -108.3



section_3 · 95.00 s - 151.00 s · clean

Labels tone_profile = clean dynamics_profile = sparse presence = background

Loudness RMS = -33.3 dB Peak = -11.8 dB Crest = 21.5 dB

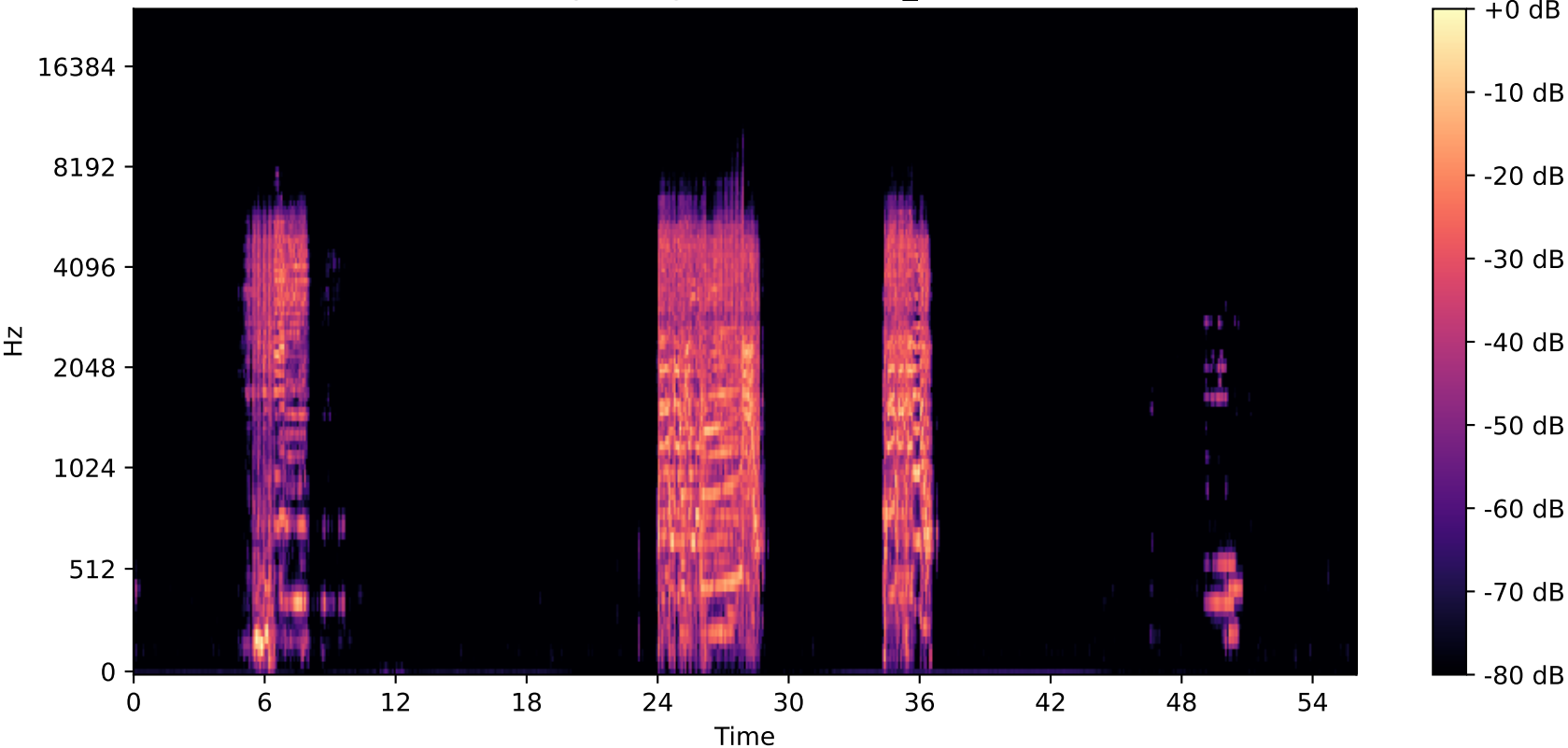
Spectrum Centroid = 2215 Hz Rolloff (85%) = 2391 Hz Flatness = 0.000

Distortion THD ≈ - % odd/even = -0.1 dB gain_character = clean (conf 0.60)

Time FX RT60 = - s delay = off (-, fb -) mod = off (-, depth -)

Band energy (dB) 80 Hz: -54.8 160 Hz: -44.0 320 Hz: -46.4 640 Hz: -40.4
1280 Hz: -38.3 2560 Hz: -43.8 5120 Hz: -58.0 10240 Hz: -107.1

Spectrogram — section_3



section_4 · 151.00 s - 211.00 s · clean

Labels tone_profile = clean dynamics_profile = sparse presence = lead

Loudness RMS = -23.1 dB Peak = -8.3 dB Crest = 14.8 dB

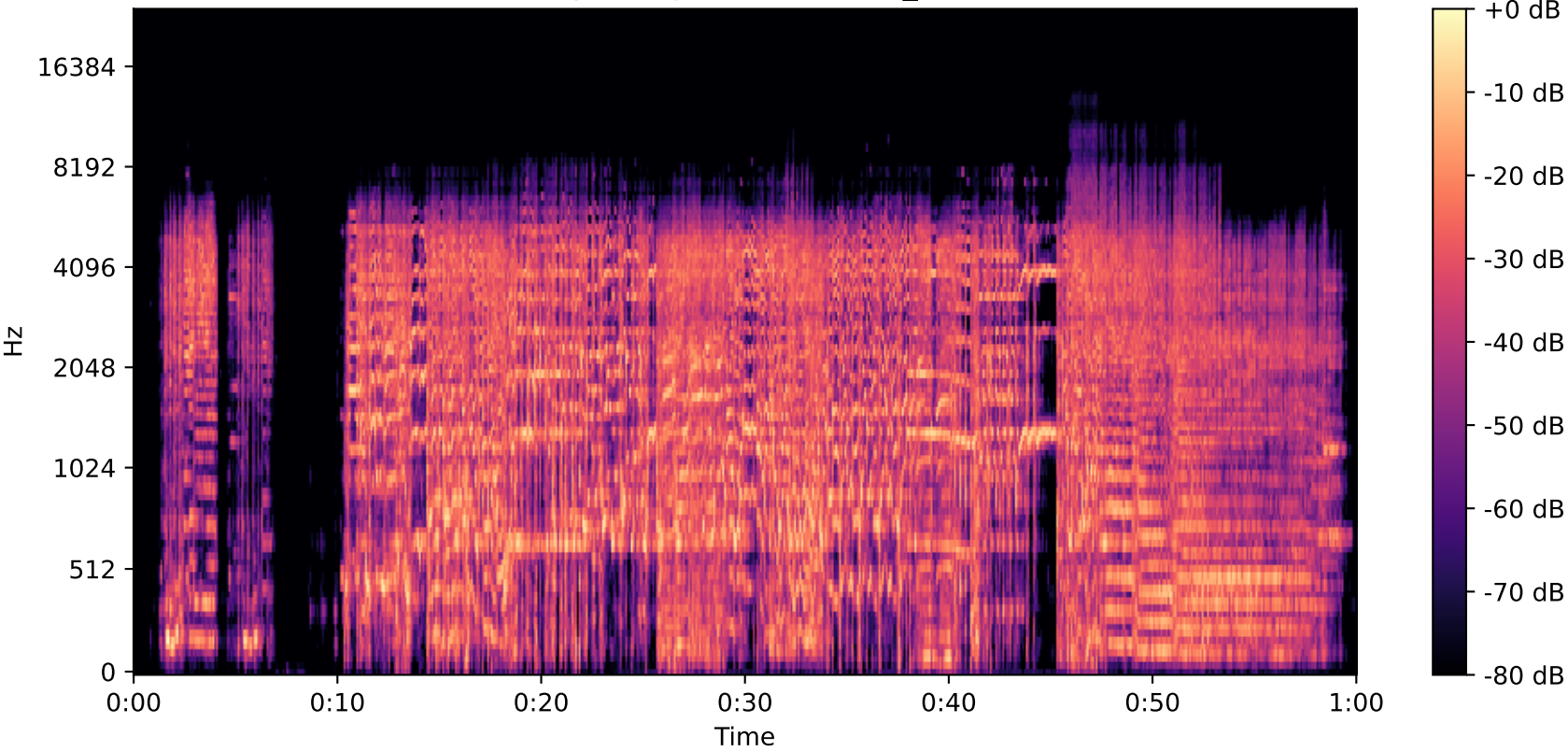
Spectrum Centroid = 2434 Hz Rolloff (85%) = 3773 Hz Flatness = 0.000

Distortion THD $\approx$ 0.0 % odd/even = +0.0 dB gain_character = clean (conf 1.00)

Time FX RT60 = - s delay = off (-, fb -) mod = off (-, depth -)

Band energy (dB) 80 Hz: -42.7 160 Hz: -38.0 320 Hz: -34.7 640 Hz: -30.3
1280 Hz: -29.0 2560 Hz: -30.3 5120 Hz: -45.8 10240 Hz: -85.9

Spectrogram — section_4



section_5 · 211.00 s - 223.00 s · clean

Labels tone_profile = clean dynamics_profile = rhythmic presence = background

Loudness RMS = -71.7 dB Peak = -43.5 dB Crest = 28.2 dB

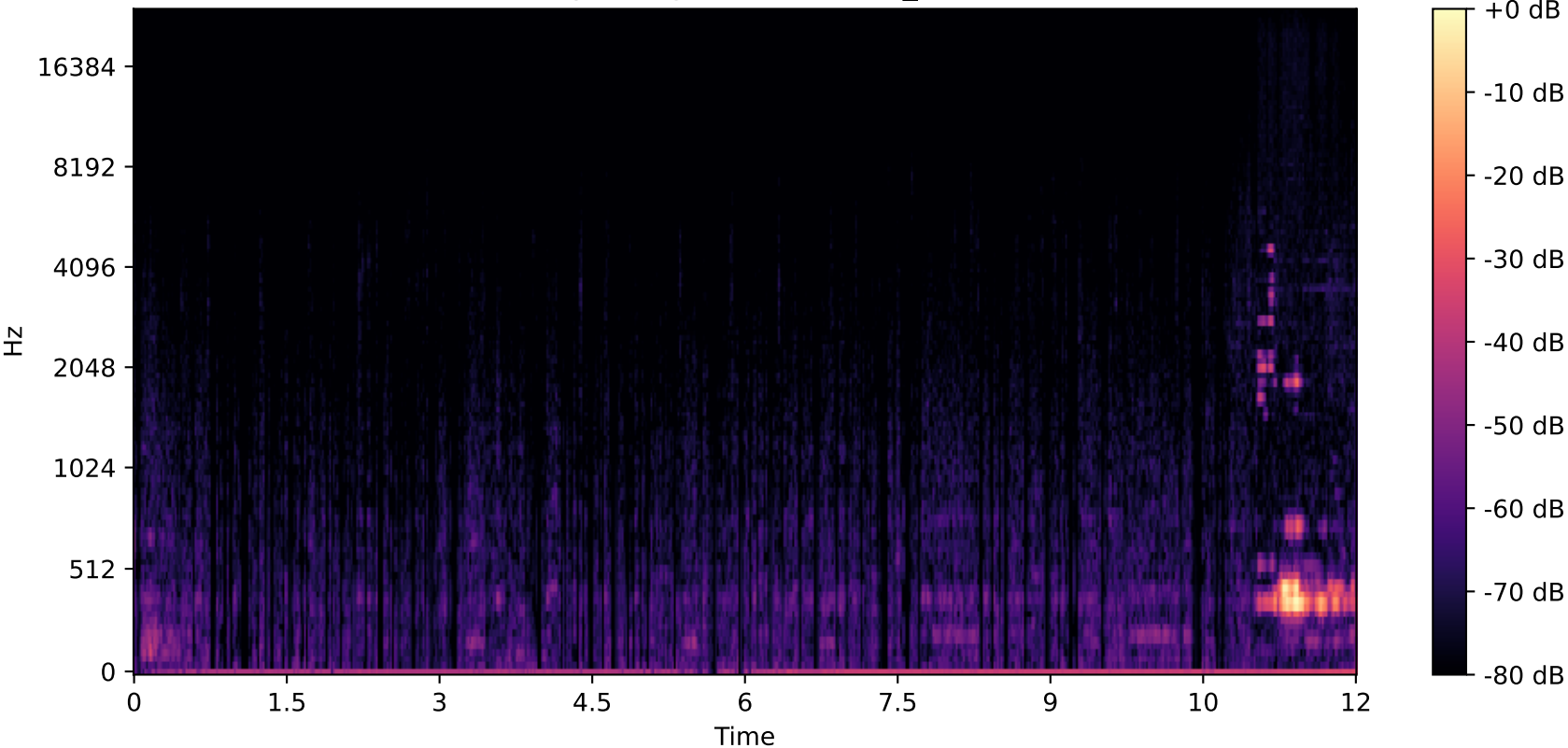
Spectrum Centroid = 3433 Hz Rolloff (85%) = 445 Hz Flatness = 0.022

Distortion THD $\approx$ 0.0 % odd/even = -11.5 dB gain_character = clean (conf 1.00)

Time FX RT60 = 1.91 s delay = off (–, fb –) mod = off (–, depth –)

Band energy (dB) 80 Hz: -109.3 160 Hz: -106.4 320 Hz: -73.1 640 Hz: -102.1
1280 Hz: -98.0 2560 Hz: -103.5 5120 Hz: -115.4 10240 Hz: -115.3

Spectrogram — section_5



section_6 · 223.00 s - 247.00 s · high_gain

Labels tone_profile = high_gain dynamics_profile = sparse presence = background

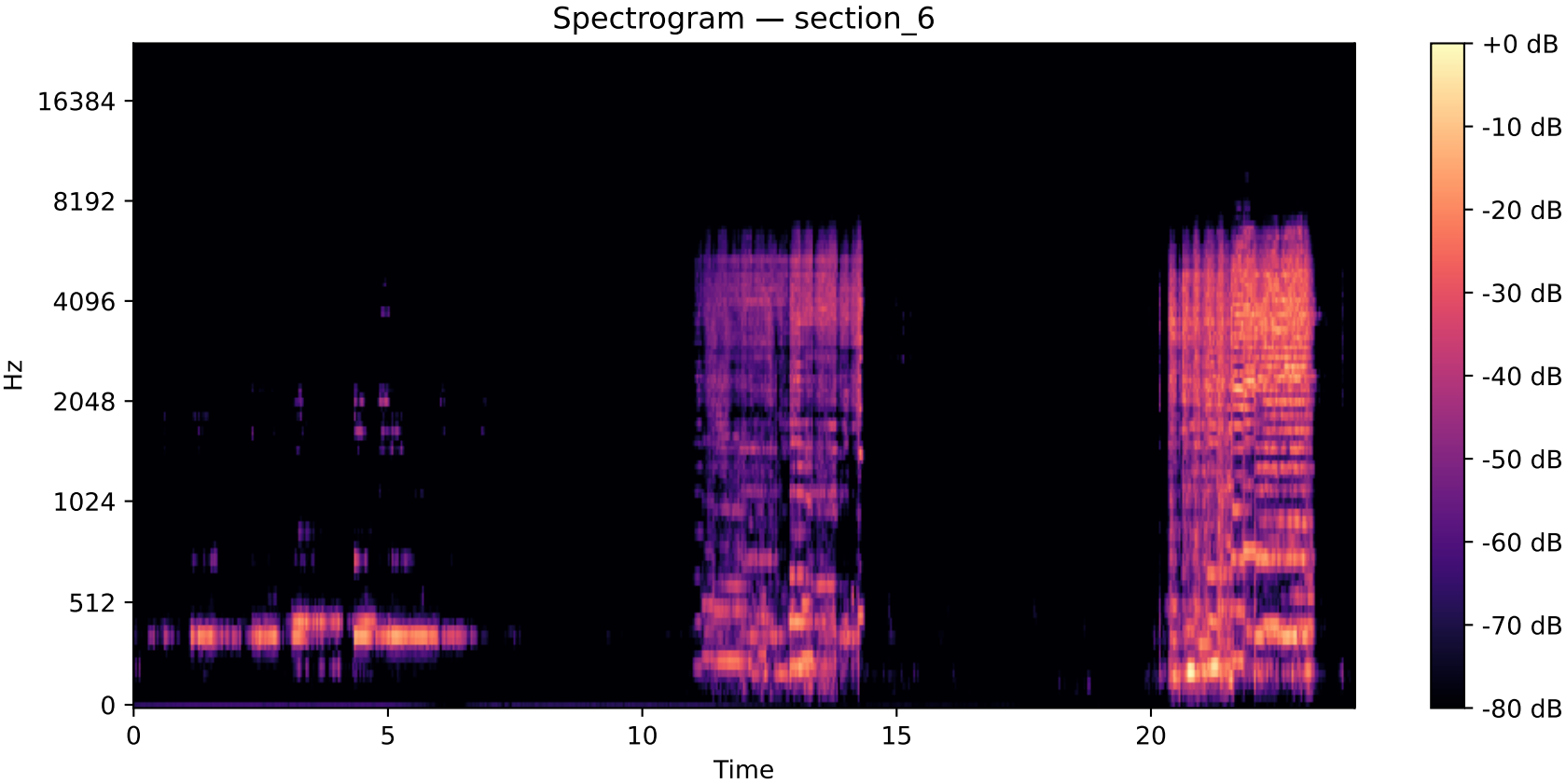
Loudness RMS = -33.9 dB Peak = -12.0 dB Crest = 21.9 dB

Spectrum Centroid = 2382 Hz Rolloff (85%) = 3938 Hz Flatness = 0.000

Distortion THD ≈ - % odd/even = -10.1 dB gain_character = high_gain (conf 0.70)

Time FX RT60 = - s delay = off (-, fb -) mod = off (-, depth -)

Band energy (dB) 80 Hz: -62.6 160 Hz: -41.6 320 Hz: -43.7 640 Hz: -49.8
1280 Hz: -42.8 2560 Hz: -38.9 5120 Hz: -53.8 10240 Hz: -107.7



section_7 · 247.00 s - 260.00 s · clean

Labels tone_profile = clean dynamics_profile = sparse presence = rhythm

Loudness RMS = -31.1 dB Peak = -11.6 dB Crest = 19.5 dB

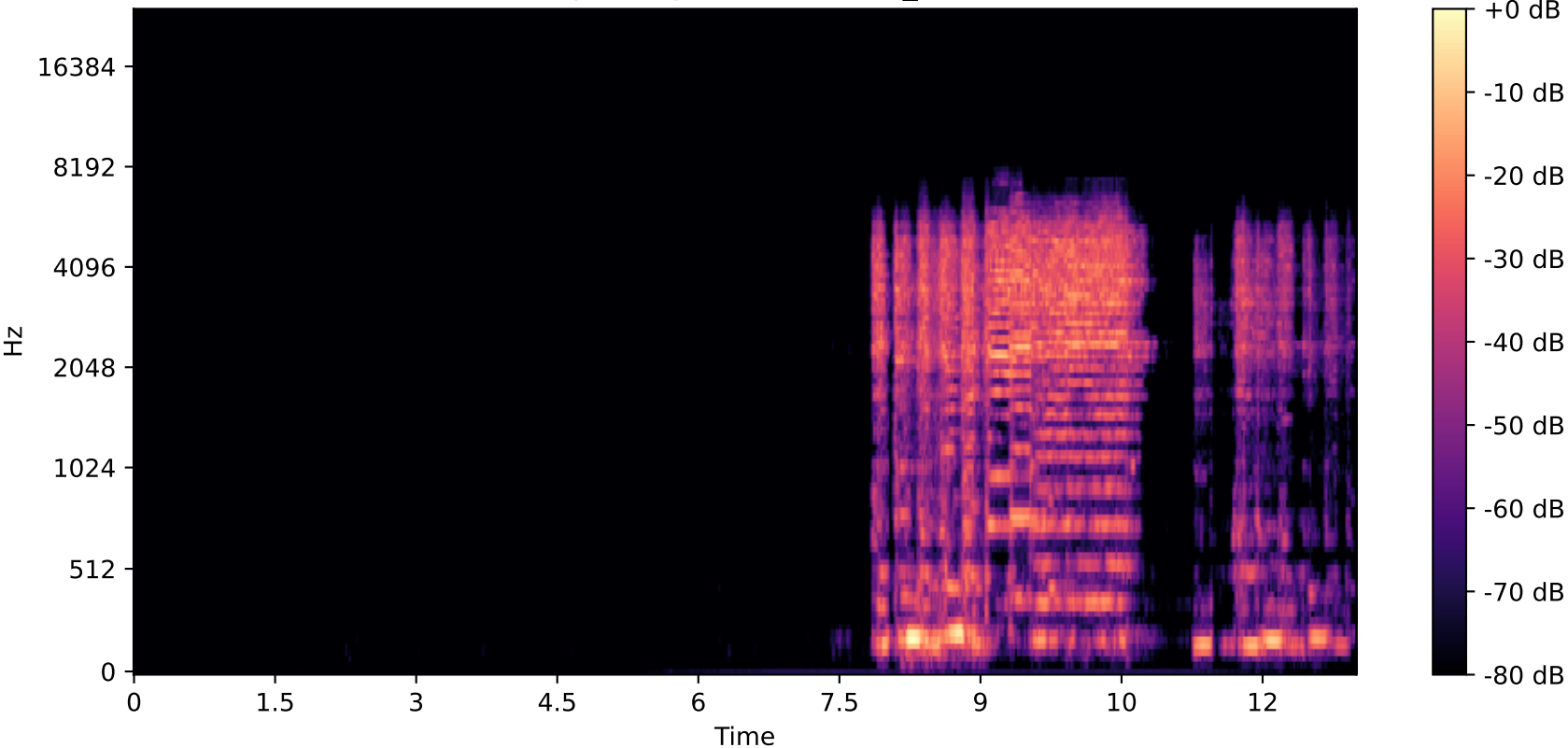
Spectrum Centroid = 2918 Hz Rolloff (85%) = 3938 Hz Flatness = 0.000

Distortion THD ≈ 0.0 % odd/even = +0.0 dB gain_character = clean (conf 1.00)

Time FX RT60 = - s delay = off (-, fb -) mod = off (-, depth -)

Band energy (dB) 80 Hz: -54.6 160 Hz: -37.0 320 Hz: -49.5 640 Hz: -47.7
1280 Hz: -39.2 2560 Hz: -36.4 5120 Hz: -51.4 10240 Hz: -108.2

Spectrogram — section_7



section_8 · 260.00 s - 279.00 s · clean

Labels tone_profile = clean dynamics_profile = rhythmic presence = background

Loudness RMS = -90.0 dB Peak = -77.2 dB Crest = 12.8 dB

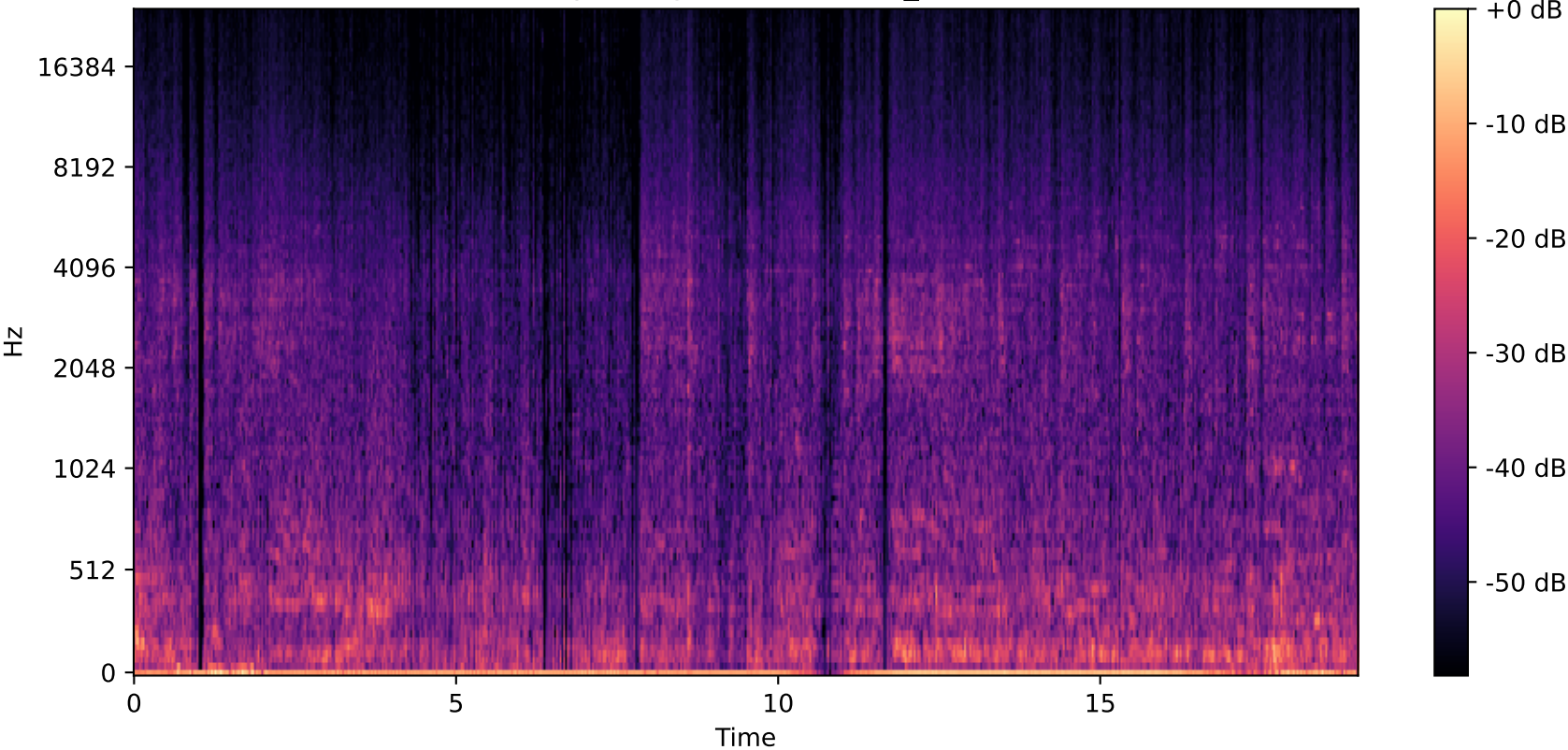
Spectrum Centroid = 5346 Hz Rolloff (85%) = 23 Hz Flatness = 0.495

Distortion THD $\approx$ 0.0 % odd/even = +0.0 dB gain_character = clean (conf 0.20)

Time FX RT60 = - s delay = off (-, fb -) mod = off (-, depth -)

Band energy (dB) 80 Hz: -107.3 160 Hz: -108.5 320 Hz: -108.5 640 Hz: -112.1
1280 Hz: -110.9 2560 Hz: -109.3 5120 Hz: -112.3 10240 Hz: -114.0

Spectrogram — section_8



section_9 · 279.00 s - 331.56 s · clean

Labels tone_profile = clean dynamics_profile = sparse presence = background

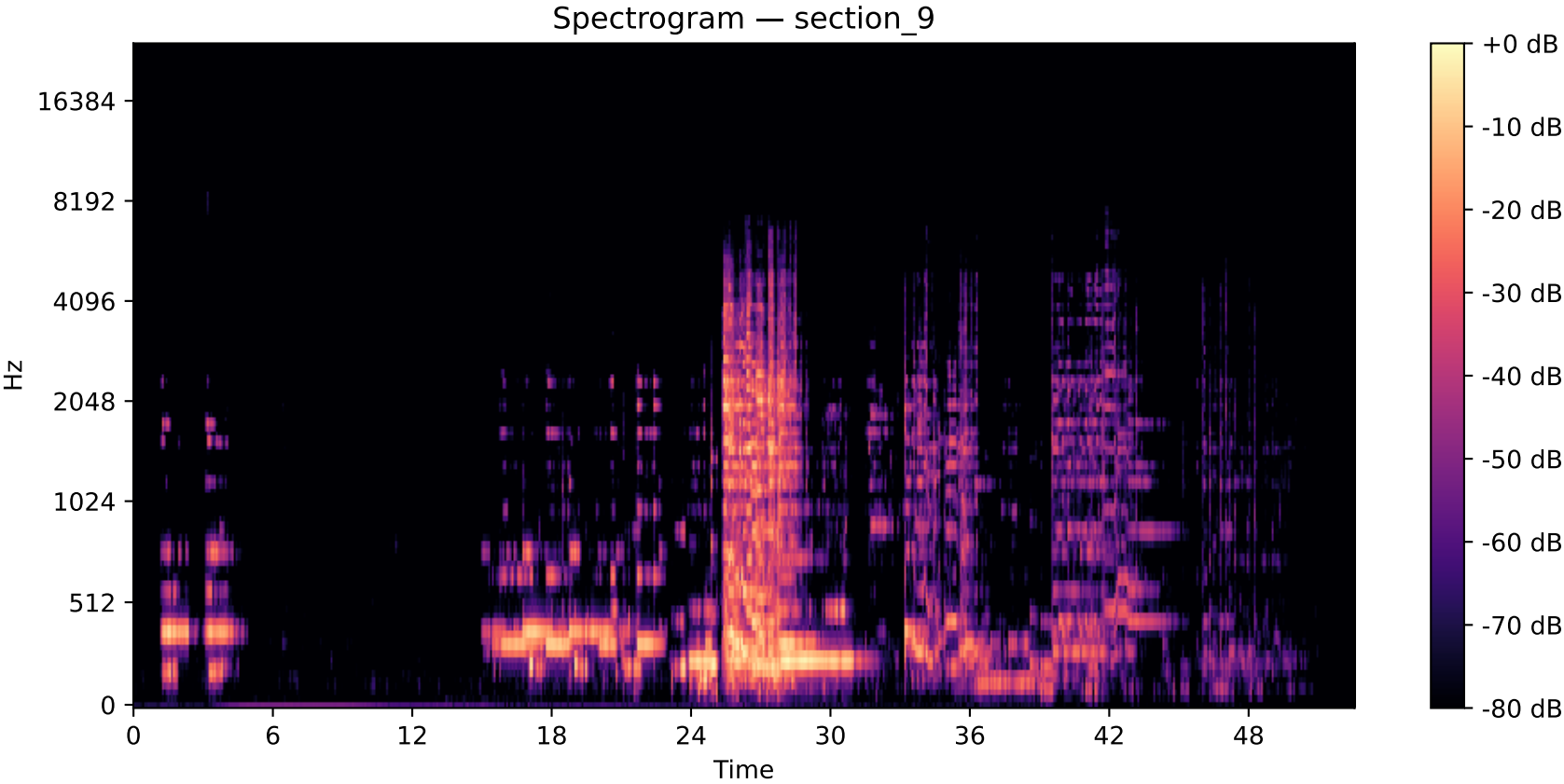
Loudness RMS = -36.4 dB Peak = -13.5 dB Crest = 22.9 dB

Spectrum Centroid = 643 Hz Rolloff (85%) = 1336 Hz Flatness = 0.000

Distortion THD $\approx$ 0.0 % odd/even = +0.0 dB gain_character = clean (conf 1.00)

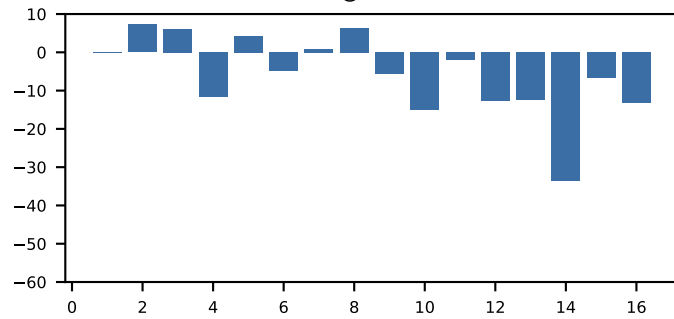
Time FX RT60 = - s delay = off (-, fb -) mod = off (-, depth -)

Band energy (dB) 80 Hz: -60.5 160 Hz: -40.7 320 Hz: -42.9 640 Hz: -52.3
1280 Hz: -46.1 2560 Hz: -56.7 5120 Hz: -82.8 10240 Hz: -107.4

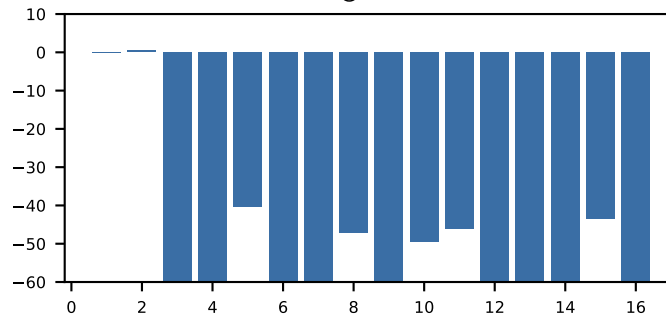


Notes

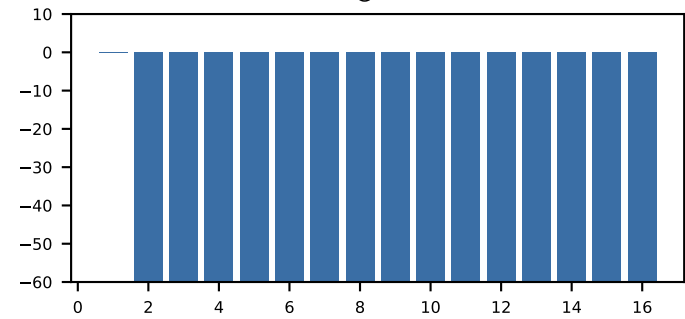
A3 @ 38.87s



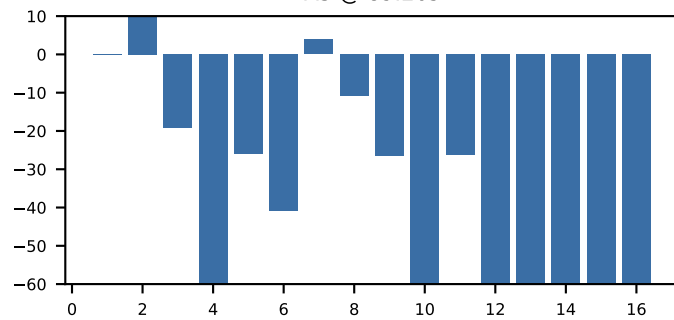
A3 @ 41.45s



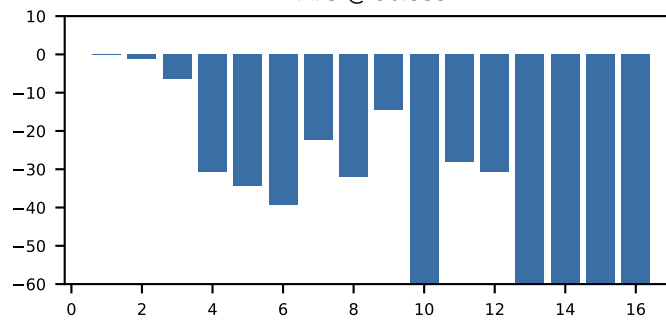
A#4 @ 75.39s



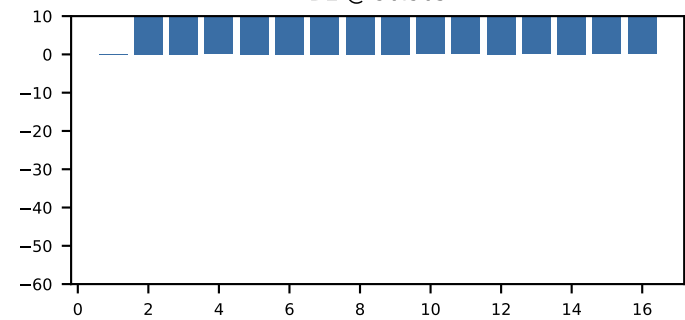
A3 @ 89.26s



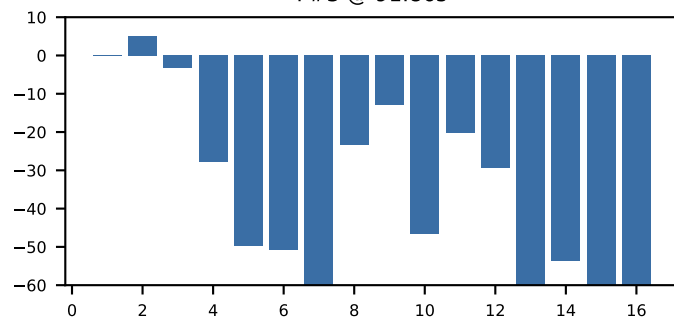
F#3 @ 90.35s



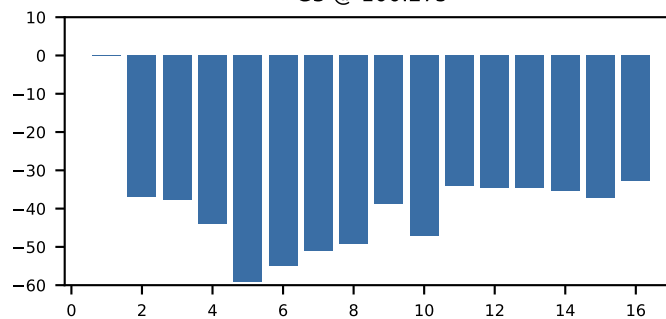
D2 @ 90.90s



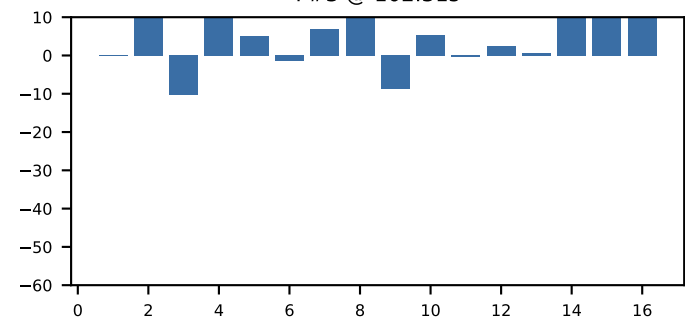
F#3 @ 91.86s



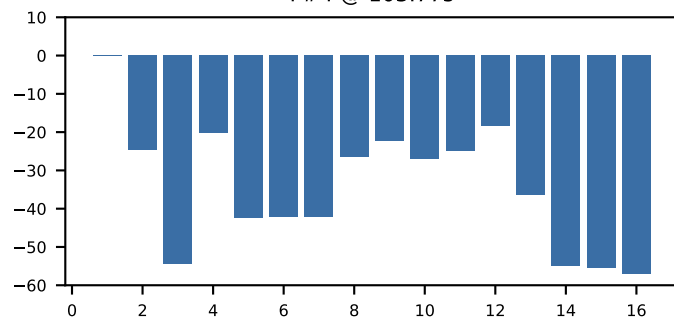
G3 @ 100.27s



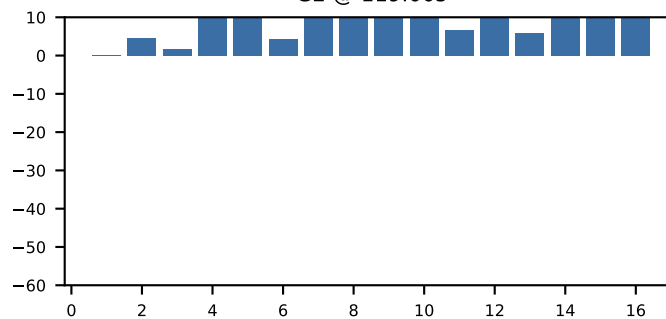
F#3 @ 102.31s



F#4 @ 103.77s



G2 @ 119.00s



A2 @ 120.96s

